Anchored residual sequence models: a bounded, penalised correction on top of a prior

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Abstract

A prior probability for a binary outcome often exists, and a recurrent network trained freely on each example's history can end up worse than it. We add the network's output to the prior's log-odds as a bounded offset whose output layer starts at zero, and penalise the squared offset itself, with a strength picked on an inner holdout. On a synthetic task with a known answer (1,000 training and 5,000 test rows), held-out log-loss falls by 0.0440 nats (SE 0.0042) against the prior alone, 0.0080 (SE 0.0024) against the same model without the penalty and 0.0695 (SE 0.0056) against an unanchored recurrent network, the one setting where it beats every arm by more than 2 SE. On ICU mortality it loses to boosted trees started from the prior by 0.0726 nats; under a boosted-tree prior it only ties the prior, while the unpenalised model is 0.1201 nats worse. Differences are paired on fixed test sets of 4,000 to 12,000 rows over 3 training subsamples; standard errors are over test examples.

1 Problem

Many binary forecasting problems come with a prior probability q_i before we train anything: a clinical severity score, an upstream model, or a probability quoted by someone else. A sequence model that reads the example's recent history can add information the prior misses. Trained without constraint, it can also do worse: it relearns what the prior already encodes, overfits at small n, and nothing anchors it to the prior under distribution shift.

On ICU mortality with 500 training stays, a recurrent network that gets the prior as an input scores a held-out log-loss of 0.6990 nats; the prior alone scores 0.3960. The usual fix is a residual on the prior's log-odds, as in residual policy learning [Silver et al., 2018, Johannink et al., 2019] and residual physics [Zeng et al., 2019]. That alone is insufficient: the same residual without a penalty scores 0.5064, still worse than the prior. Weight decay does not bound it either: Adam scales each step by the gradient's own size, so with decay 0.2 the head still sits at its bound on 62% of training rows. We want a correction that stays near the prior unless the data argue otherwise.

2 The change

The model keeps the prior as an anchor on the log-odds scale ($\operatorname{logit} q = \log \frac{q}{1-q}$) and adds a bounded, gated offset from a gated recurrent unit (GRU, a standard recurrent cell) [Cho et al., 2014]:

$$b_i = \operatorname{logit} q_i, \qquad r_i = c \tanh\!\left(w^\top h_{i,T} + \beta\right) v_i, \qquad p_i = \sigma(b_i + r_i)\,, \tag{1}$$

$$\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{BCE}(y_i, p_i) + \lambda \frac{\sum_i r_i^2}{\sum_i v_i}, \qquad \mathrm{BCE}(y,p) = -y \log p - (1-y)\log(1-p). \tag{2}$$

Preprint.

Symbol	Meaning	Symbol	Meaning
q_i, b_i	prior probability, its log-odds (clipped)	v_i	validity gate, 0 or 1
$h_{i,T}$	GRU state after the last step	c	bound on the offset (the cap)
w, β	head weights and bias, start at 0	λ	penalty strength
r_i	offset added to the prior's log-odds	σ	logistic function
p_i	predicted probability	y_i	observed label, 0 or 1

The GRU reads each value channel relative to its last observed value after a forward fill (carrying the last value forward), plus one 0/1 observation-mask channel per value channel, which is the mask of GRU-D [Che et al., 2018] without its decay. When fewer than a share 0.25 of the steps are observed we set $v_i = 0$, and the example gets the prior back. Because the head starts at zero, the untrained model equals the prior exactly, the ReZero idea [Bachlechner et al., 2021] applied to a probability. With $\lambda = 0$ our loss is the plain cross-entropy and training matches an unpenalised loop bit for bit (tested in the package).

The prior enters as a fixed offset, as in a generalised linear model [McCullagh and Nelder, 1989]. The part we study is the penalty. It acts on each example's offset r_i and not on the weights, which is where weight decay and L2-SP [Li et al., 2018] act. Penalising outputs instead of weights is known from function-space regularisation [Benjamin et al., 2019] and from distillation toward an earlier model [Li and Hoiem, 2018]; here the reference is an external prior and the penalty is the squared log-odds offset. Read as a maximum a posteriori fit (the single most probable parameter setting), λr_i^2 is a zero-mean Gaussian prior on the correction itself. The cap bounds the worst case; the penalty determines how much evidence a correction requires before it departs from the prior. We pick λ from 6 values on an inner 20% holdout of the training rows (ties go to the larger value) and refit on all training rows. Lemma 1 states what the penalty does at the population level (proof in Appendix A); a step-by-step recipe for new data is in Appendix D.

Lemma 1 (pointwise shrinkage)**.** *Fix one input with anchor b and let $\mu = \Pr(y = 1 \mid \textit{input}) \in (0, 1)$. Let $\lambda' = \lambda n / \sum_i v_i > 0$. The objective $f(r) = \mathbb{E}[\mathrm{BCE}(y, \sigma(b + r))] + \lambda' r^2$ is strictly convex, and its minimiser r^* solves $\sigma(b + r^*) + 2\lambda' r^* = \mu$. (i) $r^* = 0$ if and only if the prior is calibrated at that input, $\sigma(b) = \mu$. (ii) r^* has the sign of $\operatorname{logit}\mu - b$. (iii) $|r^*| < |\operatorname{logit}\mu - b|$ whenever $\sigma(b) \neq \mu$. With the cap, the offset is also below c in size.*

In words, the penalised correction moves toward the calibrated one and stops short of it; a calibrated prior is left alone. This is a population statement, not a promise that a fitted model never does worse than its prior.

3 Setup

Data. We use three public tasks. None of their test sets is touched by training, tuning or the choice of λ. *Synthetic*: a logistic prior on static features plus 3 random-walk channels of 24 steps with 20% missing; the true probability is known, so the best reachable log-loss is known too (0.5044 nats). *elec2* [Harries, 1999, Gama et al., 2004]: does the electricity price end above its daily average. Its prior is a logistic regression on the current row, refitted weekly on the previous 2,000 rows, and the sequence is the last 48 half-hours of 5 columns. *PhysioNet 2012* [Silva et al., 2012, Goldberger et al., 2000]: in-hospital death from an hourly grid of 36 variables over 48 hours (14.6% positive), with a logistic prior on the SAPS-I admission score, a standard ICU severity score [Le Gall et al., 1984]. We fit that prior out of fold (each row scored by a fit that never saw it); an in-sample prior teaches the residual to undo the prior's own overfit. Training sizes range from 250 to 4,000 rows.

Arms. All arms share the same windows, rows and seed. *Prior alone*. *Penalty 0*: the anchored model with $\lambda = 0$, that is, the method without our change, on the same code path. *Unanchored GRU*: the same network started at the base rate, with no cap and no penalty, and the prior's log-odds as an extra input channel. *LightGBM from prior*: gradient-boosted trees [Ke et al., 2017] whose starting score is the prior's log-odds, stopped early on the same inner holdout, fed per-variable summaries (PhysioNet) or the network's own flattened steps. *Ours*: the anchored model with the chosen λ. The fixed-λ sweep and 4 ablations run alongside. Every recurrent arm uses one small network: hidden size 8, cap $c = 2.5$, AdamW (Adam with decoupled weight decay) [Loshchilov and Hutter, 2019] at learning rate 8×10^{-3} and weight decay 0.2, one seed and one bootstrap resample (Appendix D).

Tuning budget: ours picks λ from 6 values and LightGBM its number of rounds, both on the same inner holdout; the other arms get no search.

Metric and error. We score held-out log-loss in nats, the mean negative log-probability of the true label, $\ell_i = \mathrm{BCE}(y_i, p_i)$. Against each arm a we report the paired difference and its standard error (SE), $\bar{d} = \frac{1}{n_{\text{test}}} \sum_i (\ell_i^{\text{ours}} - \ell_i^a)$ and $\mathrm{SE} = s_d / \sqrt{n_{\text{test}}}$, where s_d is the standard deviation of the per-example differences, each test example first averaged over the 3 training subsamples. Negative favours ours. Rows of elec2 depend on their neighbours, so there the SE comes from a moving-block bootstrap (resampling contiguous runs of rows) [Künsch, 1989] with one-week blocks of 336 rows and 2,000 draws. We call a difference a win only beyond 2 SE, and a tie inside one SE.

4 Results

Table 1 lists every setting. Ours beats the prior everywhere except synthetic $n = 250$, a tie (-0.0015, SE 0.0029). The margin over penalty 0 is widest where penalty 0 overfits: 0.115 nats on PhysioNet at $n = 500$, where penalty 0 even falls behind the prior. When the sequence carries strong signal (elec2 at 2,000, synthetic at 4,000) the holdout picks $\lambda = 0$, so ours and penalty 0 coincide. The unanchored GRU loses in every row, by 0.969 nats at elec2 $n = 500$ but by just 0.004 at synthetic $n = 4{,}000$, right on the boundary.

Table 1: The anchored model beats the prior and both recurrent arms almost everywhere, and boosted trees on PhysioNet beat it. Held-out log-loss (nats) and paired differences ours − arm with SE; negative favours ours; bold marks a win or loss beyond 2 SE. λ^* lists the penalty the inner holdout chose on each training subsample.

Task	n	Prior	Ours	vs prior	vs penalty 0	vs unanchored	vs LightGBM from prior	λ^*
Synthetic	250	0.5954	0.5940	−0.0015 (0.0029)	**−0.2341** (0.0079)	**−0.3308** (0.0098)	**−0.0062** (0.0030)	0.1, 0.3, 0.3
Synthetic	1,000	0.5954	0.5514	**−0.0440** (0.0042)	**−0.0080** (0.0024)	**−0.0695** (0.0056)	**−0.0294** (0.0035)	0.1, 0, 0.1
Synthetic	4,000	0.5954	0.5251	**−0.0703** (0.0047)	same model	**−0.0038** (0.0019)	**−0.0264** (0.0030)	0, 0, 0
elec2	500	0.5598	0.3291	**−0.2307** (0.0263)	−0.0173 (0.0087)	**−0.9692** (0.0918)	**−0.1738** (0.0269)	0, 0.1, 0
elec2	2,000	0.5598	0.2151	**−0.3447** (0.0205)	same model	**−0.3301** (0.0343)	**−0.2416** (0.0353)	0, 0, 0
PhysioNet	500	0.3960	0.3916	**−0.0045** (0.0007)	**−0.1148** (0.0083)	**−0.3074** (0.0170)	**+0.0471** (0.0030)	1, 1, 1
PhysioNet	2,000	0.3960	0.3811	**−0.0150** (0.0017)	**−0.0913** (0.0076)	**−0.0585** (0.0105)	**+0.0726** (0.0047)	0.3, 0.3, 0.3

Our pre-registered keep rule demands a win over all four arms, each beyond 2 SE. Only synthetic $n = 1{,}000$ passes (Figure 1a). Elec2 misses on penalty 0 alone: -0.0173 (SE 0.0087) at $n = 500$, just short of the bar, and at $n = 2{,}000$ the holdout chose penalty 0 itself. On PhysioNet the trees win by 0.047 and 0.073 nats; per-variable summaries read an irregular clinical grid better than our small GRU.

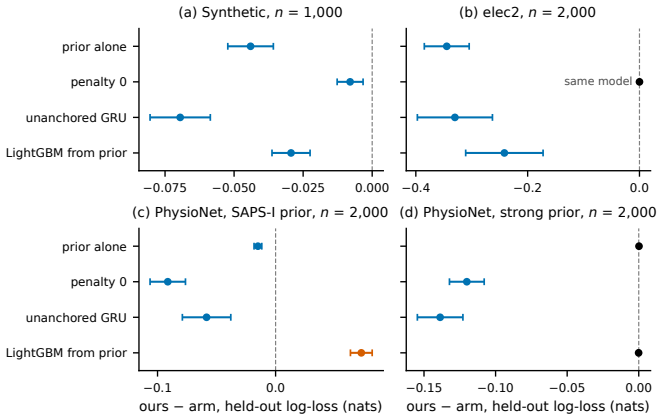


Figure 1: Ours beats every arm on synthetic $n = 1{,}000$, loses only to boosted trees on PhysioNet, and ties a strong prior. Paired difference ours − arm with 95% interval (± 1.96 SE); blue left of zero favours ours, orange right of zero favours the arm, black overlaps zero. Panel (d) uses the strong prior described in the text.

Figure 2 exposes the mechanism. At $\lambda = 0$ the model lands above the prior on PhysioNet and on synthetic $n = 250$; a moderate λ pulls it back to or below the prior. The best fixed strength drifts

with n and with the sequence's signal, and on elec2 any positive λ costs log-loss. So we choose it per split. That choice has a price. At synthetic $n = 250$ the inner holdout holds only a few dozen rows, picks a weak λ, and the row ends as a tie.

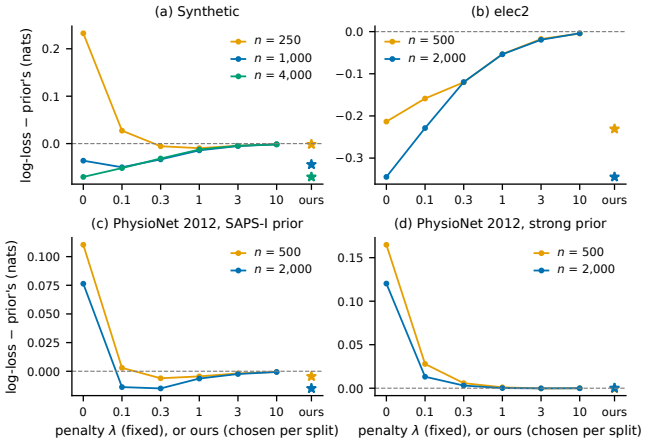


Figure 2: In these four settings the penalty turns a residual that hurts the prior into one that matches or beats it, and the best strength depends on the data. Held-out log-loss minus the prior's against a fixed λ; the star is ours with λ chosen on the inner holdout; below the dashed line beats the prior.

The penalty also steadies training. In 5-seed reruns at the smallest n, the unpenalised head saturates its cap on 44% to 90% of training rows. Moving λ from 0 to 1 shrinks the seed-to-seed spread of predicted probabilities from 0.099 to 0.014 (PhysioNet), 0.035 to 0.006 (elec2) and 0.157 to 0.020 (synthetic). Calibration follows: the ECE (expected calibration error, the mean gap between predicted and observed frequency) at PhysioNet $n = 500$ is 0.116 for penalty 0 and 0.017 for ours (reliability diagram in Appendix C).

A strong prior is the hardest test of no harm. With LightGBM on all 48-hour summaries as the prior, ours ties it (-0.0002, SE 0.0002 at 500; +0.0002, SE 0.0002 at 2,000), and penalty 0 trails ours by 0.165 and 0.120 nats. No gain, and no harm.

The method grew out of a private study. On a proprietary binary forecasting task (about a thousand held-out cases), the unpenalised residual variant degraded its prior's held-out log-loss by 0.0060 nats (paired block-bootstrap SE 0.0030). The output-space penalty removed that degradation (penalised model minus prior: -0.0001 nats, SE 0.0007), halved the seed-to-seed spread of predicted probabilities (0.008 to 0.004) and lowered ECE from 0.023 to 0.014. The residual did not beat the prior by a significant margin on that task.

5 Limits

The complete win holds in a single setting. On PhysioNet the boosted trees remain ahead, and none of the 3 variants we ran to close that gap succeeded (Appendix B). The most useful one, 10 times the weight decay, improves ours (0.3571 at 2,000) but helps the unanchored GRU more (0.3277): cap and penalty hold the model near a weak anchor. With a prior that cannot see the second day, ours is worse than the prior at $n = 500$ (+0.0017, SE 0.0008); nowhere else does it lose to its prior.

These variants are post hoc, chosen after the PhysioNet result. The arms, the 6-value sweep, the 4 ablations, the sizes and the keep rule were fixed before any run. One fix came after a run: the first elec2 tree arm saw a shorter lag window (log-loss 0.563 at 500 and 0.527 at 2,000); we report the rerun.

Our SE covers test-sample noise only. Taken over the 3 training subsamples, the SE of synthetic $n = 1{,}000$ against penalty 0 grows from 0.0024 to 0.0040, which puts that win on the 2 SE boundary. Each arm is a single network rather than the package default of 5 seeds by 3 bootstrap resamples. We dropped full-size training (compute), a GRU-D arm (library unavailable), an ablation without the mask channels and a weather task (no data access). We claim no finite-sample guarantee, no advantage over boosted trees on tabular summaries, and no novelty for the zero start or the cap.

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A Proof of Lemma 1

Given. 1) $\sigma(z) = 1/(1+e^{-z})$ has $\sigma'(z) = \sigma(z)(1-\sigma(z)) > 0$ for every real z. 2) $\frac{d}{dz}\log\sigma(z) = 1-\sigma(z)$ and $\frac{d}{dz}\log(1-\sigma(z)) = -\sigma(z)$. 3) A differentiable function with a strictly increasing derivative is strictly convex, and a point where that derivative is zero is its unique minimiser. 4) Intermediate value theorem: a continuous g with $g(a) < 0 < g(b)$ has a root strictly between a and b.

Choose: $z = b + r$, and per example the loss of Eq. (2) times n, which gives the penalty weight $\lambda' = \lambda n / \sum_i v_i$. BCE is linear in y, so the expectation gives:

$$f(r) = -\mu \log\sigma(b+r) - (1-\mu)\log\big(1-\sigma(b+r)\big) + \lambda' r^2.$$

Need to prove: $f'(r) = \sigma(b+r) - \mu + 2\lambda' r$.

$$\begin{aligned} f'(r) &= -\mu\big(1-\sigma(b+r)\big) - (1-\mu)\big(-\sigma(b+r)\big) + 2\lambda' r && \text{given 2, chain rule} \\ &= -\mu + \mu\sigma(b+r) + \sigma(b+r) - \mu\sigma(b+r) + 2\lambda' r && \text{distributive law} \\ &= \sigma(b+r) - \mu + 2\lambda' r && \text{the } \mu\sigma \text{ terms cancel} \end{aligned}$$

Then:

$$\begin{aligned} f''(r) &= \sigma'(b+r) + 2\lambda' && \text{derivative of the line above} \\ &> 0 && \text{given 1 and } \lambda' > 0 \end{aligned}$$

Therefore, by given 3, f is strictly convex and its minimiser r^* is the only root of f', so $\sigma(b+r^*) + 2\lambda' r^* = \mu$.

Choose: $\hat{r} = \operatorname{logit}\mu - b$, the calibrated offset, so $\sigma(b+\hat{r}) = \mu$.

Need to prove: $0 < r^* < \hat{r}$ if $\hat{r} > 0$, $\hat{r} < r^* < 0$ if $\hat{r} < 0$, and $r^* = 0$ if and only if $\hat{r} = 0$.

$$\begin{aligned} f'(0) &= \sigma(b) - \mu = \sigma(b) - \sigma(b+\hat{r}) && \text{substitution of } \mu \\ f'(\hat{r}) &= \sigma(b+\hat{r}) - \mu + 2\lambda'\hat{r} = 2\lambda'\hat{r} && \text{substitution of } \mu \end{aligned}$$

Case $\hat{r} > 0$**:** σ is increasing (given 1), so $f'(0) < 0$, and $f'(\hat{r}) = 2\lambda'\hat{r} > 0$.

Precondition. Given 4 needs f' continuous: f' is a sum of σ, a constant and a linear term, each continuous, so it is.

Therefore, by given 4, f' has a root in $(0, \hat{r})$, and by uniqueness it is r^*: $0 < r^* < \hat{r}$.

Same method, case $\hat{r} < 0$**:** $f'(0) > 0$ and $f'(\hat{r}) < 0$, so $\hat{r} < r^* < 0$.

Case $\hat{r} = 0$**:** $f'(0) = 0$, so $r^* = 0$. In the other direction, $r^* = 0$ gives $f'(0) = \sigma(b) - \mu = 0$, so $\sigma(b) = \mu$ and $\hat{r} = 0$.

Therefore (i) $r^* = 0$ exactly when $\sigma(b) = \mu$; (ii) r^* has the sign of $\hat{r}$; (iii) $|r^*| < |\hat{r}|$ when $\hat{r} \neq 0$. With the cap the reachable offsets are the interval $(-c, c)$. f is strictly convex, so on that interval it decreases toward r^*: the fitted offset is r^* when $|r^*| < c$ and approaches the endpoint on the side of r^* otherwise. The sign is kept and the size stays below c. □

B Ablations and variants

Table 2 removes one part of ours at a time. The validity gate and the cap barely matter on PhysioNet and the synthetic task; on elec2 the cap matters at $n = 500$ (removing it costs 0.154 nats). A random head start is mixed. Penalty 0 with 10 times the weight decay is the strongest network in several rows, which is what motivated the weight-decay variant.

Table 2: No single part of ours carries the result outside elec2; heavy weight decay is a real competitor. Paired difference ours − ablation (SE); negative means the removed part helped; bold beyond 2 SE. Last columns: ECE of the prior and of ours (mean over splits).

Task	n	no cap	random head start	no gate	penalty 0, weight decay ×10	Ours LL	ECE prior	ECE ours
Synthetic	250	+0.0013 (0.0022)	**+0.0061** (0.0023)	+0.0000 (0.0000)	**+0.0426** (0.0037)	0.5940	0.025	0.030
Synthetic	1,000	+0.0012 (0.0015)	**−0.0039** (0.0016)	+0.0000 (0.0000)	**+0.0189** (0.0024)	0.5514	0.025	0.043
Synthetic	4,000	+0.0008 (0.0005)	−0.0005 (0.0003)	+0.0000 (0.0000)	**−0.0064** (0.0018)	0.5251	0.025	0.033
elec2	500	**−0.1540** (0.0461)	**+0.0131** (0.0040)	+0.0000 (0.0000)	**+0.0300** (0.0120)	0.3291	0.094	0.030
elec2	2,000	−0.0308 (0.0161)	**−0.0157** (0.0037)	+0.0000 (0.0000)	**−0.0554** (0.0105)	0.2151	0.094	0.034
PhysioNet	500	+0.0003 (0.0005)	−0.0001 (0.0006)	−0.0000 (0.0002)	**−0.0303** (0.0055)	0.3916	0.019	0.017
PhysioNet	2,000	−0.0003 (0.0011)	+0.0014 (0.0014)	−0.0008 (0.0009)	**+0.0240** (0.0035)	0.3811	0.019	0.020

Table 3 lists the 3 post-hoc PhysioNet variants. With a strong prior that cannot see the second day (V1), ours ties it at 2,000 and is slightly worse at 500; the trees started from the prior gain a little more. With the boosted-tree prior (V2) ours ties both the prior and the trees, and penalty 0 is far worse. With 10 times the weight decay for every network (V3), the unanchored GRU overtakes ours at 2,000.

Table 3: None of the post-hoc variants beats boosted trees from the prior; under a strong prior the penalty keeps the model at the prior. Paired difference ours − arm (SE), bold beyond 2 SE.

Variant	n	Prior LL	vs prior	vs penalty 0	vs unanchored	vs LightGBM from prior	λ^*
V1 prior blind to day two	500	0.3090	**+0.0017** (0.0008)	**−0.1763** (0.0088)	**−0.2872** (0.0129)	**+0.0039** (0.0015)	1, 0.3, 10
V1 prior blind to day two	2,000	0.3090	−0.0000 (0.0001)	**−0.1262** (0.0071)	**−0.1261** (0.0084)	**+0.0057** (0.0014)	10, 10, 3
V2 boosted-tree prior	500	0.2776	−0.0002 (0.0002)	**−0.1651** (0.0082)	**−0.3711** (0.0146)	−0.0004 (0.0005)	3, 3, 10
V2 boosted-tree prior	2,000	0.2776	+0.0002 (0.0002)	**−0.1201** (0.0062)	**−0.1387** (0.0081)	−0.0001 (0.0002)	1, 10, 10
V3 weight decay ×10	500	0.3960	**−0.0096** (0.0016)	**−0.0354** (0.0050)	**−0.0170** (0.0062)	**+0.0420** (0.0030)	0.3, 10, 0.1
V3 weight decay ×10	2,000	0.3960	**−0.0389** (0.0042)	same model	**+0.0294** (0.0052)	**+0.0486** (0.0050)	0, 0, 0

C Calibration

Figure 3 shows reliability at the smallest training sizes. Penalty 0 scatters its predictions far from the diagonal, while ours stays on the prior's curve.

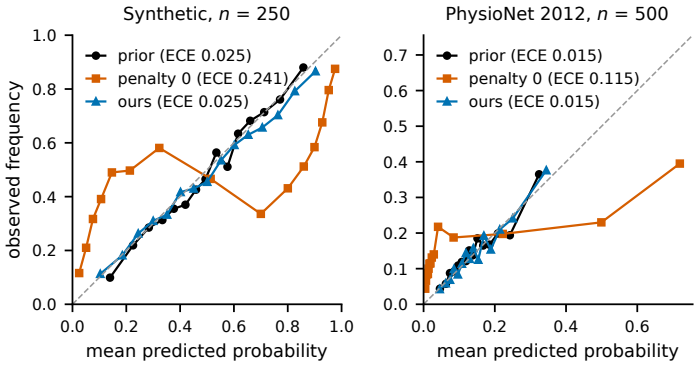


Figure 3: Without the penalty the residual is badly calibrated at small n; with it, calibration matches the prior. Reliability diagram, 15 equal-width bins of pooled test predictions. ECE in the legend is computed on the pooled predictions, so it differs slightly from the per-split means in Table 2.

D Settings and reproduction

Applying it to new data. The method needs, per example, a prior probability q_i computed without the example's label (out of fold if the prior is fitted on the same rows), a label y_i, and a window

of past values of shape steps $\times$ channels with missing values marked. Steps: clip q_i and take its log-odds; forward fill each channel, express it relative to its last observed value and add the 0/1 mask channels; fit the model of Eq. (1) with the loss of Eq. (2) for each λ in the grid on 80% of the training rows; keep the λ with the lowest log-loss on the other 20%; refit on all training rows. The package wraps this in a scikit-learn style classifier, `AnchoredGRUClassifier`, whose `fit` takes windows of shape (n, T, C) with NaN for missing values plus `prior=`q.

Benchmark settings. Network for every recurrent arm: one GRU layer, hidden size 8, cap 2.5 (package default 2.0), learning rate 8×10^{-3}, weight decay 0.2, batch 128, stride 2 over the steps, one seed and one bootstrap resample. Epochs by size: PhysioNet and elec2 100 at 500 and 30 at 2,000; synthetic 150, 60 and 20 at 250, 1,000 and 4,000. Penalty sweep $\lambda \in \{0, 0.1, 0.3, 1, 3, 10\}$, chosen on an inner 80/20 split. Trees: learning rate 0.05, 15 leaves, at most 1,000 rounds with early stopping after 50 rounds on the inner holdout. The whole benchmark took 57 minutes on 2 single-thread workers. Code at commit `4ba5111` of the `anchorgru` repository; `bench/run_all.sh` reruns the benchmark, `results/make_numbers.py` and `figures/make.py` regenerate every number and figure here. The change itself, the penalty term and the zero-started bounded head, is in `anchorgru/net/residual.py`; with $\lambda = 0$ that loss returns the plain cross-entropy, so penalty 0 is the unchanged baseline. Dependencies as pinned in `pyproject.toml` (PyTorch [Paszke et al., 2019] $\geq$ 2.1, LightGBM $\geq$ 4.0).